



NQRUST-BACKUP & RESTORE

Protect Workloads | Restore Confidently

Enterprise storage platform with unified block, file, and object storage — managed from a single web console. Scales from single-server to scale-out distributed deployments.

4-Level
Backup
Levels

Disk
+ S3
Target

Auto
Scheduling

Cross
Restore

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Introduction

What Is NQRust-Backup?

A self-hosted enterprise backup and restore platform that protects Linux, Windows, and application servers — with automated scheduling, multi-level backup strategies, and flexible storage targets.

NQRust-Backup is a centralized backup and restore platform designed for organizations that need reliable, automated data protection across mixed server environments. The platform supports **Linux and Windows** backup clients, with configurable filesets, multiple backup levels (Full, Differential, Incremental, VirtualFull), and flexible storage targets including **local disk and S3-compatible object storage** — enabling both on-site and off-site backup strategies from a single web console.

Built on a proven **Director-based architecture**, NQRust-Backup separates orchestration (Director), storage management (Storage Daemon), and client-side data collection (File Daemon) into dedicated components for reliability and scalability. The web console provides centralized visibility into job status, schedule management, storage utilization, restore operations, and analytics — all without relying on CLI for routine operations.

Core Capabilities



Multi-Level Backup

Four backup levels for optimal storage efficiency and recovery speed. Full captures the scope. Differential captures changes since last Full. Incremental captures changes since last backup of any level. VirtualFull synthesizes a full backup from existing incrementals — saving storage while maintaining full-style restore capability.



Flexible Restore

Restore multiple files or a specific file version with checksum verification. Choose the target client, backup job, and restore location. Control file replacement behavior: always overwrite, never overwrite, overwrite if older, or overwrite if newer. Restore to the original client or cross-restore to a different machine.



Local Disk + S3 Storage

Back up to local File storage for fast on-site recovery, or to S3-compatible object storage for off-site and cloud-tier protection. S3 targets integrate directly with NQRust-Storage or supported S3 endpoints. Configurable chunk size, IO threads, storage class (STANDARD_IA), and connection parameters per target.

Web Console Sections

Full backup lifecycle managed from a single authenticated web interface

Section	Capabilities
Dashboard	Overview of recent job status, storage utilization, client health, and schedule activity at a glance.
Jobs	Show all jobs, Actions, Run (manual trigger with job/client/fileset/storage/pool/level/type/priority/when), Rerun failed jobs, Timeline view with Successful/Warning/Running/Waiting/Failed status per job.
Restore	Restore multiple files with file selection tree. Restore specific file version with available versions table (modification date, size, checksum). Cross-client restore support.
Clients	Manage backup clients — Linux (backup-linux), Windows (backup-windows), and application servers (backup-nqrust-client, prxmonitor, pusdatintrans).
Schedules	Schedule definitions (WeeklyCycle, WeeklyCycleAfterBackup), Overview with config details, Scheduler Status with 7-day preview of upcoming jobs and levels.
Storages	Devices (File, S3 buckets), Object Storage (S3 target config with browser), Pools (Full/Differential/Incremental/Scratch with retention and max volume bytes), Volumes (name/pool/storage/type/status/retention/size).
Director	Central orchestration engine configuration and status monitoring.
Analytics	Backup analytics and reporting for capacity planning and compliance.

Platform Capabilities

Features That Protect Your Business

Enterprise backup capabilities — from automated scheduling to point-in-time file version recovery.



Automated Scheduling

Configurable backup cycles with Full, Differential, and Incremental rotation. WeeklyCycle example: Full Saturday, Differential alternate Saturdays, Incremental Mon-Fri. 7-day scheduler preview.



Full Backup

Complete copy of all selected files and directories. The baseline for all other backup levels. Configurable retention (1 year default) with maximum volume size controls.



Differential Backup

Captures all changes since the last Full backup. Faster than Full, restores require only Full + latest Differential. Configurable retention (3 months default).



Incremental Backup

Captures only changes since the last backup of any level. Smallest backup size, fastest execution. Configurable retention (1 month default). Ideal for daily backups.



VirtualFull

Synthesizes a full backup from existing incrementals without re-reading client data. Saves storage and network bandwidth while maintaining full-style restore capability.



Point-in-Time Restore

Restore specific file versions with modification date, file size, and checksum verification. Browse available versions and select the exact point-in-time to recover.



Multi-Platform Clients

Backup agents for Linux and Windows servers. Manage multiple clients from the central console. Cross-client restore enables recovery to a different machine than the original source.



S3 Object Storage Target

Back up directly to S3-compatible object storage. Configurable endpoint, bucket, path prefix, access keys, storage class, chunk size, IO threads, and connection timeouts. Built-in S3 browser.



Local File Storage

On-site backup to local disk for fast recovery. Volumes tracked with pool assignment, type, status, retention, and current/maximum size. Append mode for efficient space usage.



Pool Management

Organize volumes into pools: Full, Differential, Incremental, and Scratch. Per-pool retention policies, maximum volume count, and maximum volume size for controlled capacity growth.



Job Timeline & Analytics

Visual timeline showing backup job execution with status indicators (Successful, Warning, Running, Waiting, Failed). Analytics section for capacity planning and compliance reporting.



Director Orchestration

Central Director engine coordinates all backup and restore operations. Manages job scheduling, client communication, storage allocation, and catalog maintenance automatically.



Restore Deep Dive

Restore multiple files with a visual file selection tree — browse backed-up directories and select exactly what to recover. **Restore specific file version** with available versions table showing modification timestamp, size, and checksum for each version. Choose **replace behavior**: always, never, if older, if newer. Restore to original client or **cross-restore** to any other registered client. Default restore path: `/tmp/nqrust-backup-restores`.



S3 Storage Integration

Direct backup to **S3-compatible object storage** with full configuration control: endpoint host, bucket, path prefix, region, access/secret keys, HTTPS toggle, Signature v2 support, s3cmd program, **STANDARD_IA** storage class, chunk size (262MB default), IO threads (4), IO slots (2), timeout (30s), retries. Built-in **S3 Browser** to navigate backup folders in the bucket. Integrates natively with NQRust-Storage S3 engine.

Backup Level Comparison

Protocol	What It Captures	Restore Needs	Best For
Full	All selected files and directories	Full only	Weekly baseline, compliance
Differential	Changes since last Full	Full + latest Diff	Mid-week protection
Incremental	Changes since last backup (any level)	Full + all Inchr chain	Daily, minimal storage
VirtualFull	Synthesized full from incrementals	VirtualFull only	Storage-efficient full restore

Use Cases

Built for **Every** Backup Challenge **Enterprise Server Protection**

Centralized backup for mixed Linux and Windows server environments. **Automated weekly cycles** with Full, Differential, and Incremental rotation. Pool-based retention policies ensure compliance while controlling storage growth.

 Government & Regulatory Compliance

Self-hosted on-premise for **strong data sovereignty**. Point-in-time file version recovery with checksum verification for restore traceability. Configurable retention policies per backup level. Analytics for compliance reporting.

 Hybrid Backup & Off-site DR

Local disk for fast on-site recovery + **S3 object storage** for off-site disaster recovery. Back up to NQRust-Storage S3 engine or supported S3-compatible targets. Dual-target strategy helps protect data against local failures and site incidents.

 Multi-Server & Multi-Site

One Director manages backups across **multiple clients and locations**. Centralized scheduling, storage management, and restore operations. Timeline view provides fleet-wide visibility into backup health across all servers.

Under The Hood

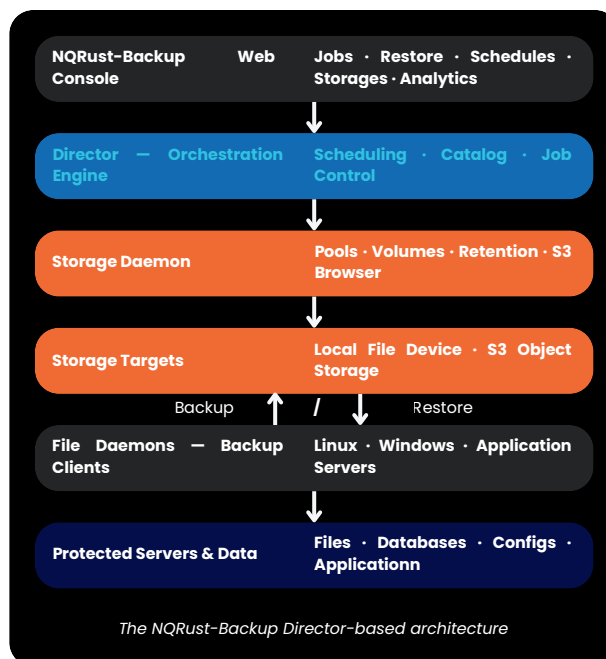
Director-based architecture — orchestration, storage, and client agents as dedicated components.

The **Director** is the central orchestration engine that manages all backup and restore operations. It maintains the job schedule, coordinates client communication, allocates storage resources, and manages the backup catalog — a database that tracks every file, version, and volume for fast restore lookups.

The **Storage Daemon** handles all data writing and reading operations. It manages local File devices and S3-compatible storage targets, organizing data into pools and volumes with configurable retention policies. Volumes can be stored on local disk (File) or in S3 buckets, with the S3 Browser providing direct visibility into backup data.

File Daemons (backup clients) run on each protected server — Linux or Windows. They collect file data and metadata during backup jobs and receive restore data during recovery operations. Multiple clients can be managed from one Director.

The **Scheduler** automates backup cycles with configurable weekly rotations. The scheduler status page shows a 7-day preview of upcoming jobs, levels, and timing — ensuring administrators always know what will run next.



Why NQRust-Backup for Your Organization

Reliable, automated data protection — from single servers to enterprise fleets.

 Proven Backup Strategy

Full + Differential + Incremental + VirtualFull backup levels. Automated weekly cycles with configurable retention. Pool-based volume management for controlled, predictable storage growth.

 Local + S3 Dual Target

Back up to local disk for fast recovery and to S3 for off-site disaster protection. Integrates natively with NQRust-Storage. Dual-target strategy helps protect data against single points of failure.

 Granular Point-in-Time Recovery

Restore individual files or entire directories. Select specific file versions by date, size, and checksum. Cross-client restore to any registered machine. Flexible file replacement policies.

 Predictable Costs, Long-Term Value

Web console manages jobs, schedules, storage, clients, and restores — CLI not required for routine operations. Timeline view and analytics provide fleet-wide backup health at a glance.